BE (E&TC)	PLC SCADA and Automation LAB	
Experiment No: 7	Control the speed of AC 3Φ motor using VFD	**Pages**

AIM: To Control the speed of AC 3Φ motor using VFD.

THEORY:

Various automation processes in the industry need control of AC induction motors using AC drives. Presented here is a robust system for switching on/off, varying the speed and direction of rotation of an industrial 3 Phase Induction Motor using VFD and PLC. We use here Delta AC motor drive for its operation.A simple control panel is wired using an Allen Bradley PLC for demonstration. An extended IntouchWonderware SCADA can be developed also.

An electrical motor is an electromechanical device that converts electrical energy into mechanical energy. In the case of 3-phase AC operation, the most-widely-used motor is the 3 Phase Induction Motor as this type of motor does not require any starting device, being a self-starting motor.

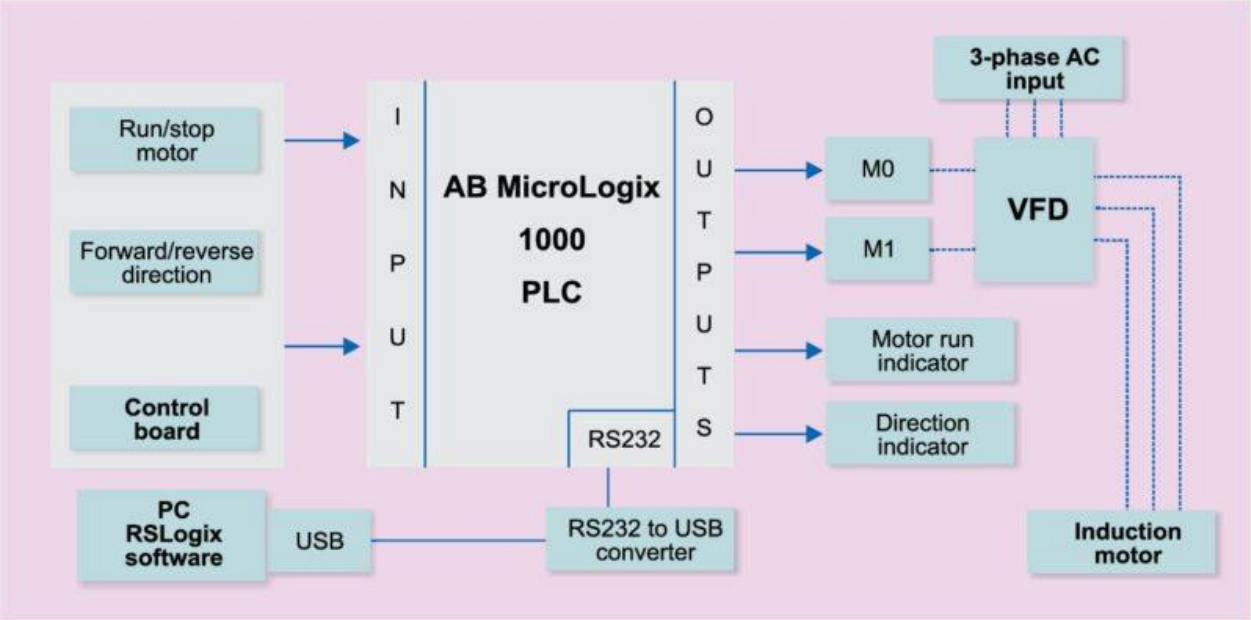


Fig. 1: Entire system layout

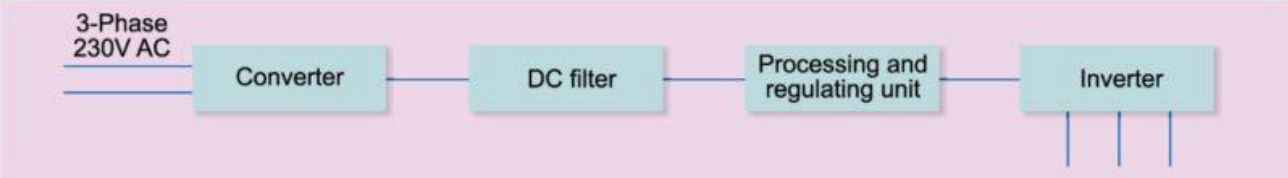


Fig. 2: Block diagram of a drive

Drives:

- Often in the industry, need arises for controlling the speed of a 3 Phase Induction Motor. Delta's AC motor drives are able to efficiently control motor speed, improve machine automation and save energy. Each drive in its variable frequency drive (VFD) series is designed to meet specific application needs.
- AC drives accurately control torque, smoothly handle increased load and provide numerous custom control and configuration operating modes. A VFD can be used to vary speed, direction and other parameters of a 3-phase motor. We use the 2-wire method for controlling the speed and direction of the motor.

Working of a VFD:

- The first stage of a VFD is the converter, which comprises six diodes, which are similar to check valves used in plumbing systems. These allow current to flow in only one direction; the direction shown by the arrow in the diode symbol. For example, whenever A-phase voltage (voltage is similar to pressure in plumbing systems) is more positive than B- or C-phase voltages, that diode opens and allows current to flow.
- When B phase becomes more positive than A phase, B-phase diode opens and A-phase diode closes. The same is true for the three diodes on the negative side of the bus. Thus, we get six current pulses as each diode opens and closes. This is called a 6-pulse VFD, which is the standard configuration for current VFDs.
- We can get rid of AC ripple on DC bus by adding a capacitor. A capacitor operates in a similar fashion to a reservoir or accumulator in a plumbing system. It absorbs AC ripple and delivers smooth DC voltage.
- The diode bridge converter that converts AC to DC is sometimes just referred to as a converter. The converter that converts DC back to AC is also a converter, but to distinguish it from the diode converter, it is usually referred to as an inverter. It has become common in the industry to refer to any DC-to-AC converter as an inverter.
- When we close one of the top switches in the inverter, that phase of the motor is connected to the positive DC bus and voltage on that phase becomes positive. When we close one of the bottom switches in the converter, that phase is connected to the negative DC bus and becomes negative. Thus, we can make any phase on the motor positive or negative at will and can thus generate any frequency that we want. So we can make any phase positive, negative or zero.

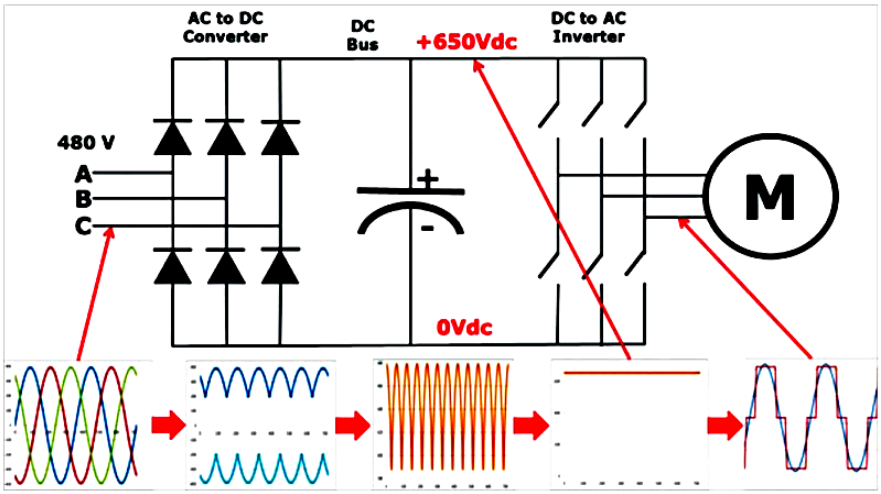


Fig. 3: Circuit model of a VFD

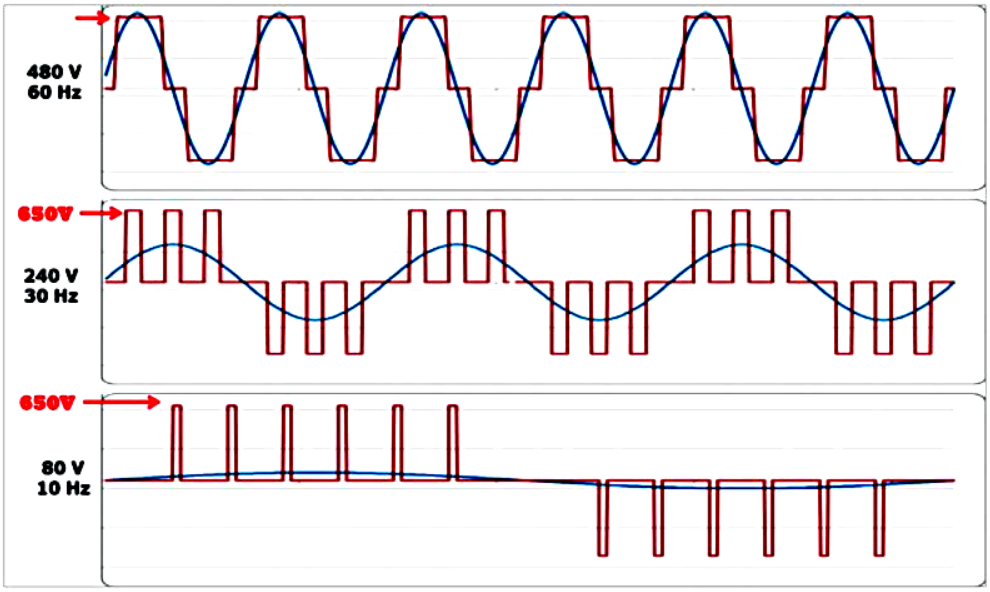


Fig. 4: Waveforms at different operating frequencies and average voltages

- Notice that, the output from the VFD is a rectangular waveform. VFDs do not produce a sinusoidal output. This rectangular waveform would not be a good choice for a general-purpose distribution system, but is perfectly adequate for a motor.
- If we want to reduce motor frequency, we simply switch the inverter output transistors more slowly. But if we reduce frequency, we must also reduce voltage in order to maintain V/Hz ratio. Pulse width modulation (PWM) does this.
- Imagine, we could control the pressure in a water line by turning the valve on and off at high speed. While this would not be practical for plumbing systems, it works very well for VFDs.
- Notice that, during the first half-cycle, voltage is on half the time and off the rest. Thus, the average voltage is half of 480V, that is, 240V. By pulsing the output, we can achieve any average voltage on the output of the VFD.

Selection of Delta VFD-M as AC drive:



Fig. 5: Delta VFD-M

- Delta VFD-M is a sensor-less vector micro AC drive. Its compact design is ideal for small- and medium-horsepower applications. M drive is designed to provide an ultra-low-noise operation and includes several innovative technologies that reduce interference.
- This drive can have numerous applications like a packing machine, dumpling machine, treadmill, temperature/humidity-control fan for agriculture and aquaculture, mixer for food processing, grinding machine, drilling machine, small-size hydraulic lathe, elevator, coating equipment, small-size milling machine, robot arm of an injection machine (clamp), wood machine (two-side woodworking planer), edge-bending machine, elasticiser and so on.

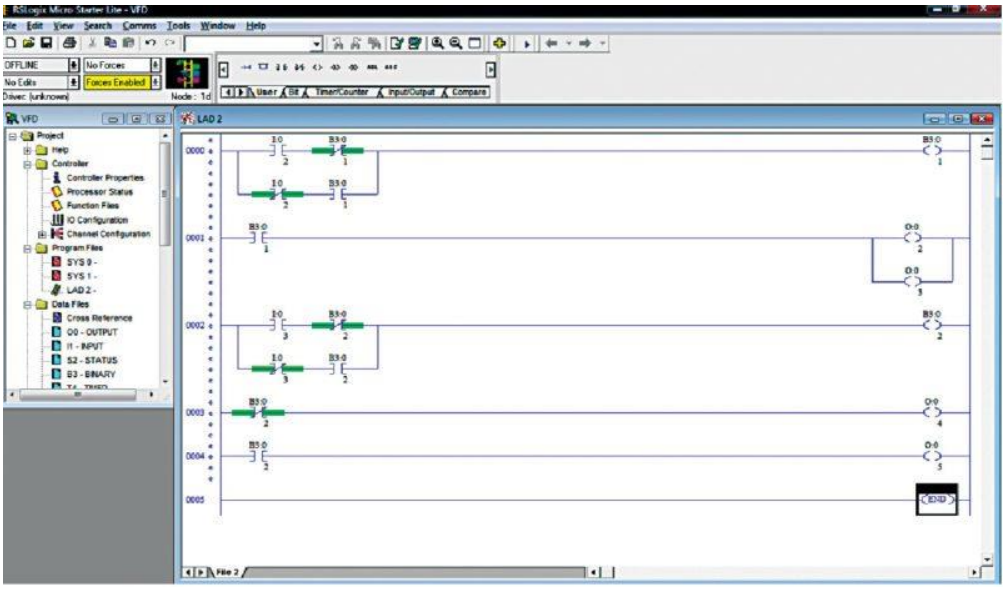


Fig.: Ladder program for control

Using the PLC:

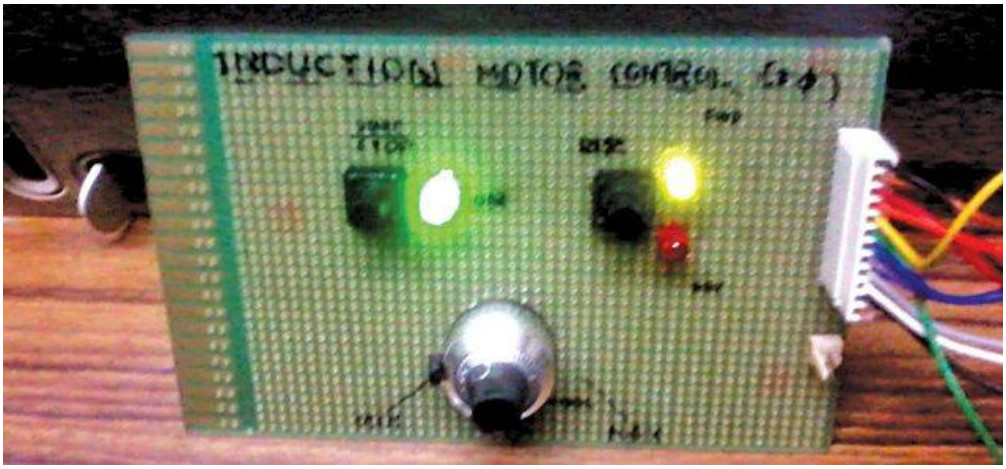


Fig. 10: Control board for motor control operation

Programmable logic controllers (PLCs) support digital input/output very effectively. So, a PLC can also be used to control the operation of a VFD, hence to finally control the connected 3 Phase Induction Motor.

CONCLUSION:

QUESTIONS:

1. How does a variable frequency drive work?
2. What is the difference between VFD and Soft Starters?
3. What are the different types of breaking methods used in a VFD?

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